

Daffodil International University
Faculty of Science & Information Technology
Department of Computer Science & Engineering
Quiz-1 Examination

Course Code:	CSE411	Course Title:	Artificial Intelligence
Section:	RE_A(3C)	Time:	50 Minutes Total Marks: 15

Question 1 (5 Marks): Define Artificial Intelligence. Explain the four views of AI: thinking humanly, thinking rationally, acting humanly, and acting rationally with suitable examples. Also write a simple Prolog rule for the statement:

“X is the grandfather of Z if X is the parent of Y, Y is the parent of Z, and X is male.”

Question 2 (5 Marks): A smart classroom cleaning robot can detect dust, identify obstacles, avoid students, clean dirty areas, and return to the charging station when its battery is low. Explain this system as an intelligent agent by identifying its environment, sensors, actuators, performance measure, and suitable agent type.

Question 3 (5 Marks): Solve the 8-Queen problem by placing 8 queens on the following 8 x 8 chessboard so that no two queens attack each other horizontally, vertically, or diagonally. Also mention one AI search technique that can be used to solve this problem. The red Q marks fixed queens that must not be moved.

8-Queen Visual Board

	A	B	C	D	E	F	G	H
1	Q							
2					Q			
3								
4						Q		
5								
6								
7								
8								

Instruction: Write Q in eight valid squares. No two queens may be in the same row, same column, or same diagonal.